CORE S115 Fall 2026: Regression demo

One predictor

Generate some data

```
x = c(1:100)
y1 = 10 + x*0.5 + rnorm(100, 0, 5)
y2 = 20 + x*-2 + rnorm(100, 0, 5)
y3 = 40 + rnorm(100, 0, 5)

dat= data.frame(x=x,
                y1=y1,
                y2=y2,
                y3=y3)
```

Comprehension check In the simulated data, what do you think is the relationship between:

- x and y1
- x and y2
- x and y3

Fit regression model

Here is code that fits regression models for each of the three y values.

```
fit1 = lm(y1 ~ x, data = dat)
summary(fit1)
##
## Call:
## lm(formula = y1 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.5568  -3.3019  -0.3729   3.2991  12.0285
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 11.42513    1.10993   10.29   <2e-16 ***
## x            0.47128    0.01908   24.70   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 5.508 on 98 degrees of freedom
## Multiple R-squared:  0.8616, Adjusted R-squared:  0.8602
## F-statistic:   610 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
fit2 = lm(y2 ~ x, data = dat)
summary(fit2)
##
## Call:
## lm(formula = y2 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.0679  -4.6418   0.3442   3.6003  13.2571
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 20.12784    1.04833    19.2   <2e-16 ***
## x           -2.00115    0.01802  -111.0   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.202 on 98 degrees of freedom
## Multiple R-squared:  0.9921, Adjusted R-squared:  0.992
## F-statistic: 1.233e+04 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
fit3 = lm(y3 ~ x, data = dat)
summary(fit3)
##
## Call:
## lm(formula = y3 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.3796  -3.2780   0.0217   3.3426  10.6502
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 39.767010   1.001789  39.696   <2e-16 ***
## x           -0.001299   0.017222  -0.075     0.94
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.971 on 98 degrees of freedom
## Multiple R-squared:  5.803e-05,  Adjusted R-squared:  -0.01015
## F-statistic: 0.005687 on 1 and 98 DF,  p-value: 0.94
```

Comprehension check What do you notice about the coefficient values and the p-values? Does this make sense? Why is the intercept different in each case?

Two predictors

Here is code that will generate y values that are a combination of two predictors, x1 and x2.

```
x1 = c(1:100)
x2 = rep(c(0, 1), times=50)

y1 = 50 + x1*0.5 + x2*-20 + rnorm(100, 0, 5)
y2 = 50 + x1*-2 + x2*0.01 + rnorm(100, 0, 10)

dat2= data.frame(x1=x1,
                 x2=x2,
                 y1=y1,
                 y2=y2)
```

Comprehension check In the simulated data, what do you think is the relationship between:

- x1, x2 and y1
- x1, x2 and y2

Fit regression model

Here is code that fits regression models with two predictors.

```
fit4 = lm(y1 ~ x1 + x2, data=dat2)
summary(fit4)
##
## Call:
## lm(formula = y1 ~ x1 + x2, data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0152  -3.3591   0.0496   2.5581  12.2102
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  50.41267    1.05279   47.88   <2e-16 ***
## x1            0.49127    0.01631   30.12   <2e-16 ***
## x2          -20.25926    0.94167  -21.51   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.708 on 97 degrees of freedom
## Multiple R-squared:  0.9329, Adjusted R-squared:  0.9315
## F-statistic:   674 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
fit5 = lm(y2 ~ x1 + x2, data=dat2)
summary(fit5)
##
## Call:
```

```
## lm(formula = y2 ~ x1 + x2, data = dat2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -37.397  -7.286  -0.553   6.222  28.106
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 50.98446    2.46845  20.654   <2e-16 ***
## x1          -2.03366    0.03824 -53.176   <2e-16 ***
## x2           1.02581    2.20792   0.465    0.643
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.04 on 97 degrees of freedom
## Multiple R-squared:  0.9668, Adjusted R-squared:  0.9662
## F-statistic:  1414 on 2 and 97 DF,  p-value: < 2.2e-16
```

Comprehension check What do you notice about the coefficient values and the p-values? Does this make sense? Specifically consider the coefficient and pvalue for x2 in fit5.

Interactions

Here is code that will generate y values that are a combination of two predictors, x1 and x2, and where the two predictors interact.

```
x1 = c(1:100)
x2 = rep(c(0, 1), times=50)

y1 = 50 + x1*0.5 + x2*-20 + x1*x2*1 + rnorm(100, 0, 5)
y2 = 50 + x1*-2 + x2*10 + x1*x2*-0.01 + rnorm(100, 0, 5)

dat2= data.frame(x1=x1,
                 x2=x2,
                 y1=y1,
                 y2=y2)
```

```
fit6 = lm(y1 ~ x1 + x2 + x1:x2, data=dat2)
summary(fit6)
##
## Call:
## lm(formula = y1 ~ x1 + x2 + x1:x2, data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.5627  -3.1293  -0.5171   2.9406  15.4106
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  48.65687    1.58905   30.62  < 2e-16 ***
```

```
## x1            0.53643    0.02752   19.49  < 2e-16 ***
## x2          -18.15797    2.26422   -8.02 2.58e-12 ***
## x1:x2         0.94398    0.03893   24.25  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.617 on 96 degrees of freedom
## Multiple R-squared:  0.9766, Adjusted R-squared:  0.9759
## F-statistic:  1337 on 3 and 96 DF,  p-value: < 2.2e-16
```

```
fit7 = lm(y2 ~ x1 + x2 + x1:x2, data=dat2)
summary(fit7)
##
## Call:
## lm(formula = y2 ~ x1 + x2 + x1:x2, data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.9467  -3.1987  -0.4493   3.5637  15.7928
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 51.13648    1.51923  33.659  < 2e-16 ***
## x1          -1.99665    0.02632 -75.875  < 2e-16 ***
## x2          11.13976    2.16473   5.146 1.41e-06 ***
## x1:x2       -0.06153    0.03722  -1.653    0.102
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.37 on 96 degrees of freedom
## Multiple R-squared:  0.992,  Adjusted R-squared:  0.9918
## F-statistic:  3968 on 3 and 96 DF,  p-value: < 2.2e-16
```

Factors

Categorical variable with two categories

When data is categorical R automatically assigns one of the predictors as the baseline – specifically the predictor which is alphabetically first. Consider the following case.

```
x = rep(c('Instrumental', 'Noise'), times=50)
x_num = ifelse(x=='Noise', 0, 1)

y = 65 + x_num*10 + rnorm(100, 0, 5)


dat3 = data.frame(x=x,
                  y=y)
```

```
fit8 <- lm(y~x, data=dat3)
summary(fit8)
##
## Call:
## lm(formula = y ~ x, data = dat3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -14.653  -3.348  -0.047   3.584  11.658
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  75.1144     0.7202 104.290  < 2e-16 ***
## xNoise       -9.9246     1.0186  -9.744 4.36e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.093 on 98 degrees of freedom
## Multiple R-squared:  0.4921, Adjusted R-squared:  0.4869
## F-statistic: 94.94 on 1 and 98 DF,  p-value: 4.36e-16
```

Comprehension check Why is the coefficient negative? What does this tell you about what is being considered a "baseline"? Does this make sense?

Categorical variable with more than two categories

And if the variable has more than two categories (or "levels") then it assigns the first alphabetical as the "baseline" and computes a coefficient comparing the baseline to each of the others.

```
x = rep(c('Pop', 'Instrumental', 'Noise'), times=50)
x_num = ifelse(x=='Noise', 0, 1)

y = 65 + x_num*10 + rnorm(150, 0, 5)


dat4 = data.frame(x=x,
                  y=y)
```

```
fit9 <- lm(y~x, data=dat4)
summary(fit9)
##
## Call:
## lm(formula = y ~ x, data = dat4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.0633  -3.2869   0.0663   3.4606  12.9933
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  75.367992   0.733223  102.79   <2e-16 ***
## xNoise      -11.877521   1.036933  -11.45   <2e-16 ***
## xPop         -0.009856   1.036933   -0.01    0.992
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.185 on 147 degrees of freedom
## Multiple R-squared:  0.5432, Adjusted R-squared:  0.537
## F-statistic:  87.4 on 2 and 147 DF,  p-value: < 2.2e-16
```

Comprehension check Why is one of the coefficients negative and one positive? What does this tell you about what is being considered the "baseline"? Does this baseline make sense?

Changing what is assigned as the baseline

The following code explicitly changes the ordering of the categories, and therefore what is considered baseline

```
dat5 = dat4 %>%
  mutate(x = factor(x, levels=c('Noise', 'Instrumental', 'Pop')))
```

```
fit10 <- lm(y~x, data=dat5)
summary(fit10)
##
## Call:
## lm(formula = y ~ x, data = dat5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.0633  -3.2869   0.0663   3.4606  12.9933
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    63.4905     0.7332   86.59   <2e-16 ***
## xInstrumental  11.8775     1.0369   11.45   <2e-16 ***
## xPop           11.8677     1.0369   11.45   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.185 on 147 degrees of freedom
## Multiple R-squared:  0.5432, Adjusted R-squared:  0.537
## F-statistic:  87.4 on 2 and 147 DF,  p-value: < 2.2e-16
```

Comprehension check What could you do if you wanted to set Pop as the baseline?

Extra comprehension check (its ok if you don't fully get this!) You will notice that only the intercept in `fit8` matches the number we were adding. The other two intercepts were off by about 10. Why do you think this is? *Hint: look at* ***x_num*** *and what role it plays in the data generation!*